

Q1:

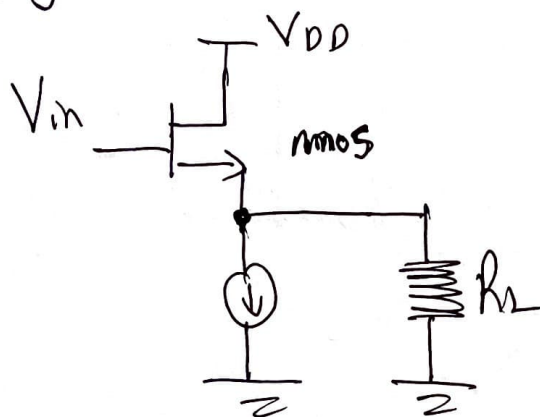
①

A class A nmos source follower output stage is biased using an ideal constant current source.

Quiescent Current = $I_Q = 10 \text{ mA}$

Load Resistance: $R_L = 500 \Omega$

Supply Voltage: $V_{DD} = 10 \text{ V}$



* What is the maximum undistorted peak output current?

$$I_{o, \text{peak}, \text{max}} = I_Q = 10 \text{ mA}$$

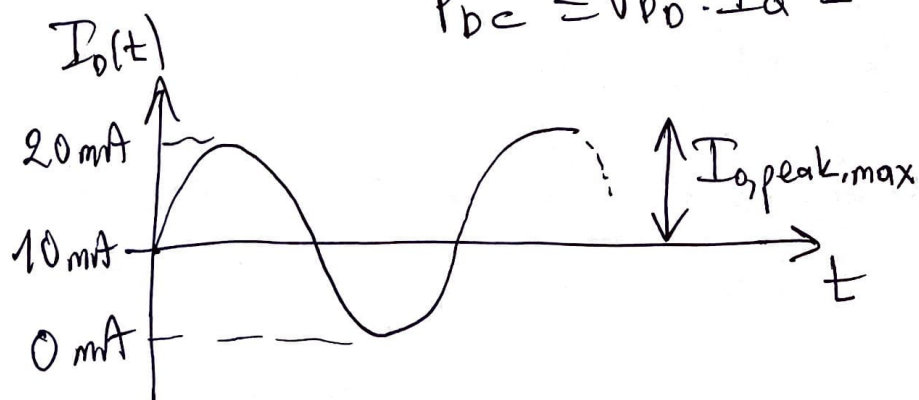
* What is the maximum undistorted peak output

voltage across the load?

$$V_{o, \text{peak}, \text{max}} = I_{o, \text{peak}} \cdot R_L = 10 \text{ mA} \cdot 500 \Omega = 5 \text{ V}$$

* What is the DC power dissipated in the MOSFET in quiescent conditions?

$$P_{DC} = V_{DD} \cdot I_Q = 10 \cdot 10 \text{ mA} = 100 \text{ mW}$$



$$I_{D, \text{max}} = I_Q + I_{o, \text{peak}, \text{max}}$$

$$I_{D, \text{max}} = \frac{10}{500} = 20 \text{ mA}$$

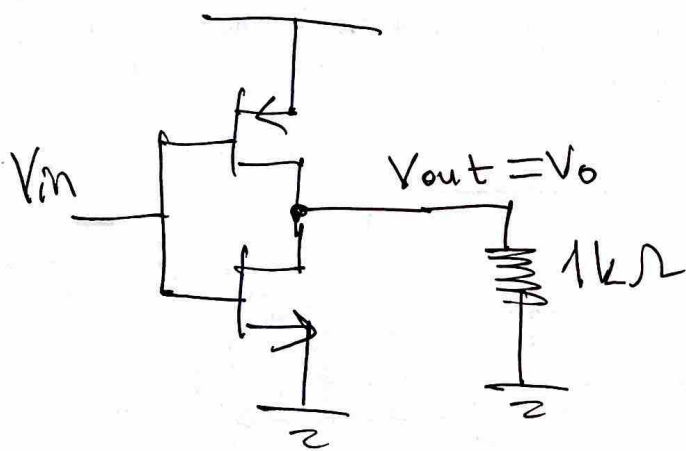
Q2:) A Class B CMOS push-pull output stage (2)
drives a resistive load.

$$V_o(t) = 4 \cdot \sin(\omega t)$$

$$R_L = 1\text{ k}\Omega \quad V_{DD} = \pm 5\text{ V} \quad V_{o, \text{peak}} = 4\text{ V}$$

a) * Calculate the peak output current.

$$I_{o, \text{peak}} = \frac{V_{o, \text{peak}}}{R_L} = \frac{4}{1000} = \boxed{4\text{ mA}}$$



b) * During which half-cycle does the nmos conduct?
 - nmos conducts during the positive half cycle of the input signal
 - pmos conducts during the negative half-cycle

of the input signal.

c) Calculate the RMS output current.

$$I_{o, \text{rms}} = \frac{I_{o, \text{peak}}}{\sqrt{2}} = \frac{4}{\sqrt{2}} = \boxed{2.83\text{ mA}}$$

d) Calculate the average power delivered to the load.

$$P_L = I_{o, \text{rms}}^2 \cdot R_L = (2.83\text{ mA})^2 \cdot 1000 = \boxed{8\text{ mW}}$$

e) Explain crossover distortion and determine the dead-zone width in the output voltage of

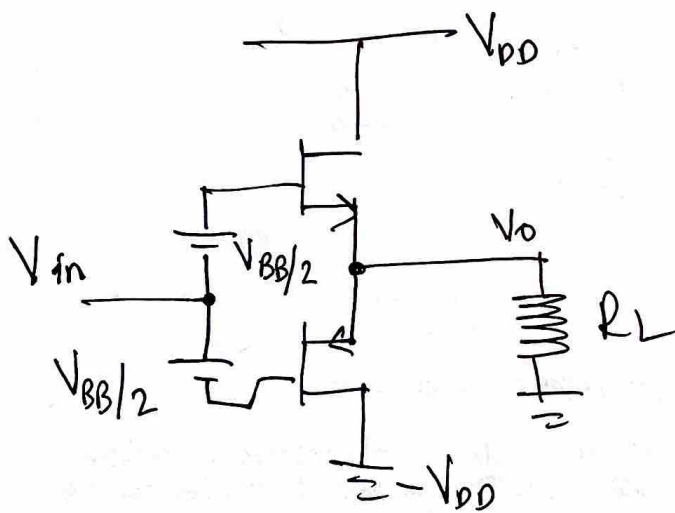
$$V_{th, n} = 0.7\text{ V}$$

$$V_{th, p} = -0.7\text{ V}$$

In Class B operation, the nmos turns on when $V_{GS} > V_{thn}$ and the pmos turns on when $V_{SG} > |V_{thp}|$.
 Around zero input voltage, both transistors are off causing crossover distortion.

$$\text{Dead-Zone width} \approx 2V_{th} = 1.4V$$

Q3:) Determine the required biasing in a MOSFET class AB output stage.



MOSFET Class-AB Stage

$$V_{DD} = 10V, R_L = 20\Omega$$

Transistors are matched

$$K = 0.2 A/V^2$$

$$|V_{th}| = 1.0V$$

The quiescent drain current is to be 20% of the load current when $V_o = 5.0V$.

$$\text{for } V_o = 5.0V$$

$$i_L = \frac{5}{20} = 0.25A$$

$$\text{for } I_Q = 0.25 \times 0.2 = 0.05A$$

when $V_o = 0V$, we have

$$I_{OQ} \approx 0.05 = K \left(\frac{V_{BB}}{2} - |V_{th}| \right)^2$$

$$= 0.2 \left(\frac{V_{BB}}{2} - 1 \right)^2$$

$$V_{BB}/2 \geq 1.50V$$

The input voltage for V_o positive is

$$V_m = V_o + V_{GSn} - \frac{V_{BB}}{2}$$

$$V_{GSn} = \sqrt{\frac{i_{Dn}}{K}} + V_{th}$$

$$= \sqrt{\frac{0.25}{0.2}} + 1 = 2.12V$$

$$V_{SGp} = V_{BB} - V_{GSn} \approx 3.2V = 0.88V$$

which means M_p cut-off

$$i_{Dn} = i_L$$

$$V_m = 5 + 2.12 - 1.5 = 5.62V$$